

DATA ANALYSIS INTERNSHIP

Weather and Climate Pattern Analysis

A Ten-Year Daily Record, 2008 to 2017

Data Quality · Seasonality · Trend Testing · Insights · Power BI
Dashboard

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1. Executive summary

The brief asks for a cleaned dataset, exploratory analysis, visualisations, an interactive dashboard and clear insights. All five are delivered.

The file holds **3,271 daily observations** across 22 columns, from 1 February 2008 to 25 June 2017. It has **no missing cells and no duplicate rows**, and every internal consistency rule passes. A routine audit would declare it clean.

Three findings say otherwise, and none of them is visible to a missing-value count.

First, nearly a third of the wind record is a placeholder. The pair *west, 41 km/h* occurs on 1,027 days, 31.4% of the file and 32 times the next most frequent direction-and-speed combination. It covers every day of 2008 and 2009 and 80% of 2010. Both values are legal in their columns, so no completeness or validity check reaches it. Computed on the full file the prevailing wind appears westerly on 43.6% of days; on the 2,244 reliable days it is westerly on 17.7%.

Second, the record is not the calendar. 162 days in the span carry no row at all. A missing row leaves no trace in a missing-value count, because there is no cell to be empty. Whole months are absent, and the observed-day count per calendar month ranges from 240 to 310, a 29% spread. Every monthly figure in this report is therefore given twice, as a raw total and normalised per observed day, and a conclusion is accepted only where the two agree.

Third, the file cannot answer the spatial question. The scenario describes patterns across cities and countries. This file carries no city, country or station column. It is a single-station daily record, so no spatial comparison is attempted. What the file supports well is the temporal question: seasonality, extremes and trend.

The accompanying Power BI dashboard presents the indicators and the charts. Its six indicator cards were verified against the data and are correct. Four statements in its insight panel were written before this analysis was complete and do not match the findings above; they are listed in section 8 and will be corrected in the next revision.

2. The data

Table 1. The file as supplied

Property	Value
Daily observations	3,271
Columns	22
First date	2008-02-01
Last date	2017-06-25
Calendar days in the span	3,433
Missing cells	0
Duplicate rows	0
Calendar days with no row	162 (4.7%)
Location column	none

The measured quantities are the daily minimum and maximum temperature, rainfall, evaporation, sunshine hours, wind gust direction and speed, wind speed and humidity

at 9 am and 3 pm, pressure at both hours, cloud cover in oktas at both hours, and two derived rain flags.

On the date format. The `Date` column is text. Read as `m/d/Y` it parses completely; read as `d/m/Y` it leaves 1,974 rows unparsed, which is 60% of the file. A single conversion call with the wrong assumption would discard those rows without raising anything, so the format was tested rather than assumed.

3. Data quality

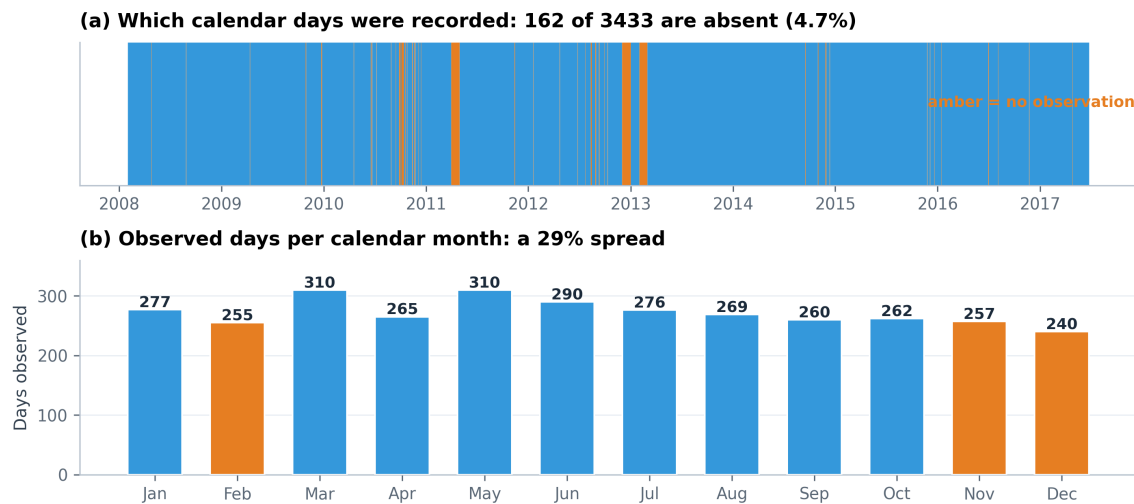


Figure 1. Calendar coverage. Panel (a) marks in amber every day of the span that carries no observation; the three widest bands are whole months. Panel (b) shows the consequence: the number of observed days differs by 29% between the best and worst covered calendar month, so a raw monthly total compares unequal samples.

Table 2. Consistency rules, all of which pass

Rule	Breaches
Maximum temperature below minimum temperature	0
Humidity outside 0 to 100%	0
Pressure outside 870 to 1085 hPa	0
Negative rainfall	0
Negative wind speed	0
RainToday disagrees with rainfall above 1 mm	0
Cloud cover outside the 0 to 8 okta scale	1

On the single cloud value. One record carries a cloud cover of 9. The okta scale runs from 0, a clear sky, to 8, complete overcast. A value of 9 is used in some archives as a code meaning the sky was obscured, by fog or by falling snow, which is not the same as overcast. It was set to missing rather than to 8, because substituting 8 would invent an observation that was never made.

3.1 The constant-pair test

A consistency rule cannot catch a value that is valid but invented. The test applied here looks for a direction-and-speed pair that repeats more often than weather allows. Wind

gust is the natural candidate, because direction and speed come from one instrument: when it fails, both are filled together.

Table 3. Share of each year carrying the pair *west, 41 km/h*

Year	Days observed	Carrying the pair	Share
2008	333	333	100.0%
2009	359	359	100.0%
2010	334	266	79.6%
2011	333	9	2.7%
2012	320	8	2.5%
2013	337	16	4.7%
2014	357	4	1.1%
2015	362	13	3.6%
2016	361	18	5.0%
2017	175	1	0.6%
Total	3,271	1,027	31.4%

The pattern is unambiguous. The pair is not scattered through the record as a coincidence would be: it fills the first two years completely, fades through 2010 and then all but disappears. This is a backfill of the period before the gust instrument began reporting, and **wind gust is treated as unusable before 2011** throughout this report.

Why no other check finds it. West is a legal direction and 41 km/h is a legal speed, inside the observed range of 17 to 96. The cells are populated, so a completeness audit sees nothing. The values are in range, so a validity rule sees nothing. Only the improbable repetition of the exact pair gives it away.

On RainToday. The flag agrees with the rainfall column on all 3,271 rows. It is therefore derived from rainfall rather than independently recorded, and it carries no information the rainfall column does not already hold.

4. Cleaning

Nine derived measures are added: the daily mean temperature, the diurnal range, mean humidity, mean wind speed, mean pressure, mean cloud, the calendar fields, and the season. **No row was deleted**; 22 columns become 35.

Table 4. One decision per problem, with the reason for it

Item	Count	Decision and reason
Date format	3,271	Parsed as m/d/Y after testing both conventions. The alternative discards 60% of the rows in silence.
Cloud cover of 9	1	Set to missing, not to 8. The code means the sky was obscured, which is a different observation from overcast.
Absent calendar days	162	Not filled. Interpolating a day's weather would invent 162 observations. The gap is carried forward and every monthly figure is normalised for it instead.
Seasons	all	Southern-hemisphere convention: December to February is summer. The warmest month is January and the coolest is July, so the northern convention would label the hottest quarter winter and invert every seasonal conclusion.
Incomplete years	2	2012 has 320 observed days and 2017 stops in June with 175. Both are excluded from year-on-year comparison and kept for everything else.

5. Indicators

Table 5. The record in eleven figures

Observations	3,271 days
Mean temperature	18.94 °C
Highest temperature	45.80 °C
Lowest temperature	4.30 °C
Mean humidity	61.47%
Mean wind speed	17.19 km/h
Strongest gust	96 km/h
Total rainfall	10,932 mm
Rainy days	849 (26.0% of days)
Mean pressure	1017.2 hPa
Mean sunshine	7.17 hours

Table 6. The five extremes in the record

Event	Date	Value
Hottest day	2013-01-18	45.8 °C
Coldest day	2010-06-30	4.3 °C
Wettest day	2015-04-21	119.4 mm
Strongest gust	2014-06-28	96 km/h
Most humid day	2011-10-20	100%

6. Patterns

6.1 The seasonal cycle

6.2 Rainfall

June receives 5.90 mm per observed day against 1.73 mm in September, a factor of 3.4. Autumn is the wettest season at 4.00 mm per day, and spring the driest at 2.31 mm.

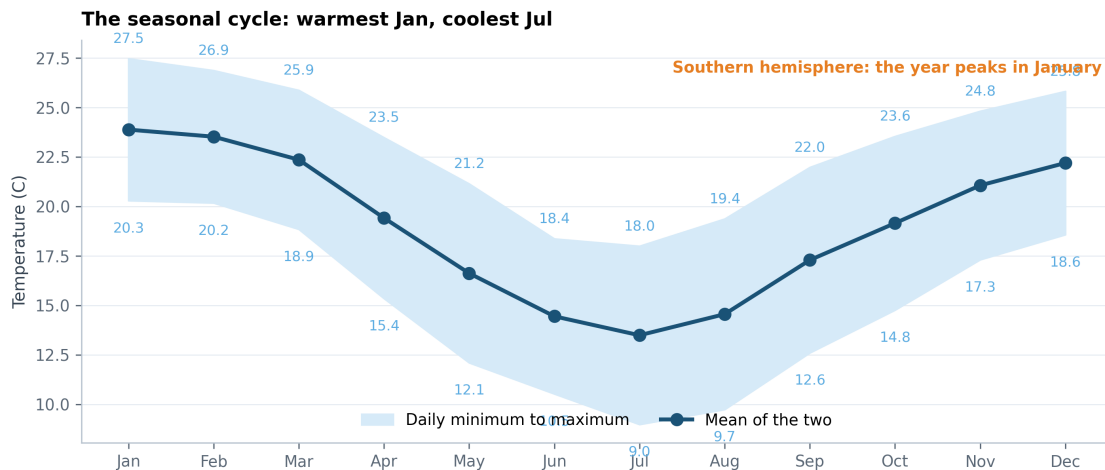


Figure 2. The annual temperature cycle. The band spans the mean daily minimum to the mean daily maximum; the line is the mean of the two. The year peaks in January and troughs in July, which is the southern-hemisphere signature and the reason the seasonal convention was set accordingly.

Table 7. The four seasons, southern-hemisphere convention

Season	Days	Mean temp	Rain per day	Humidity
Summer (Dec to Feb)	772	23.25 °C	3.31 mm	63.8%
Autumn (Mar to May)	885	19.48 °C	4.00 mm	64.4%
Winter (Jun to Aug)	835	14.18 °C	3.63 mm	60.5%
Spring (Sep to Nov)	779	19.17 °C	2.31 mm	56.8%

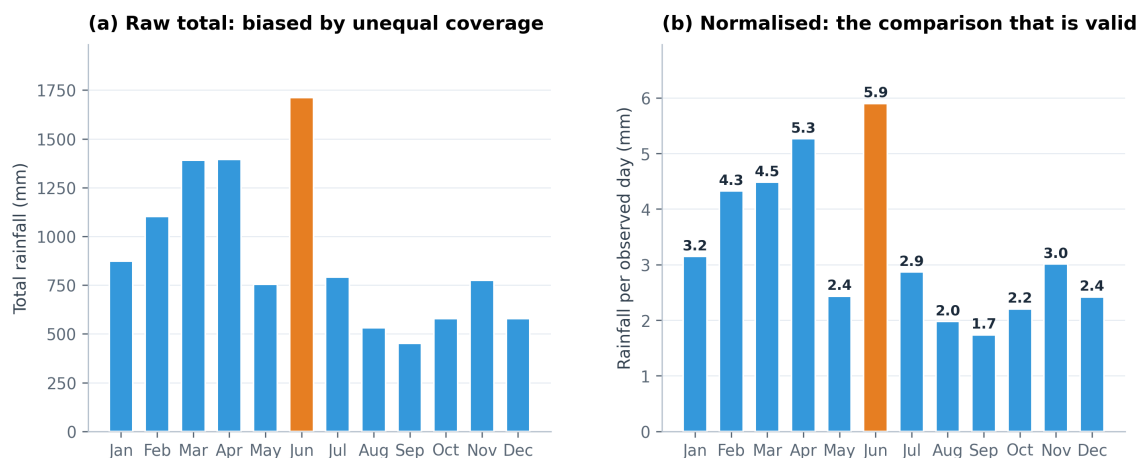


Figure 3. Rainfall by calendar month, ranked two ways. Panel (a) is the raw total, which compares months observed for different numbers of days. Panel (b) divides by the observed days and is the comparison that is valid. The two rankings agree, so June is genuinely the wettest month and September the driest; the agreement is the evidence, not an assumption.

6.3 Wind

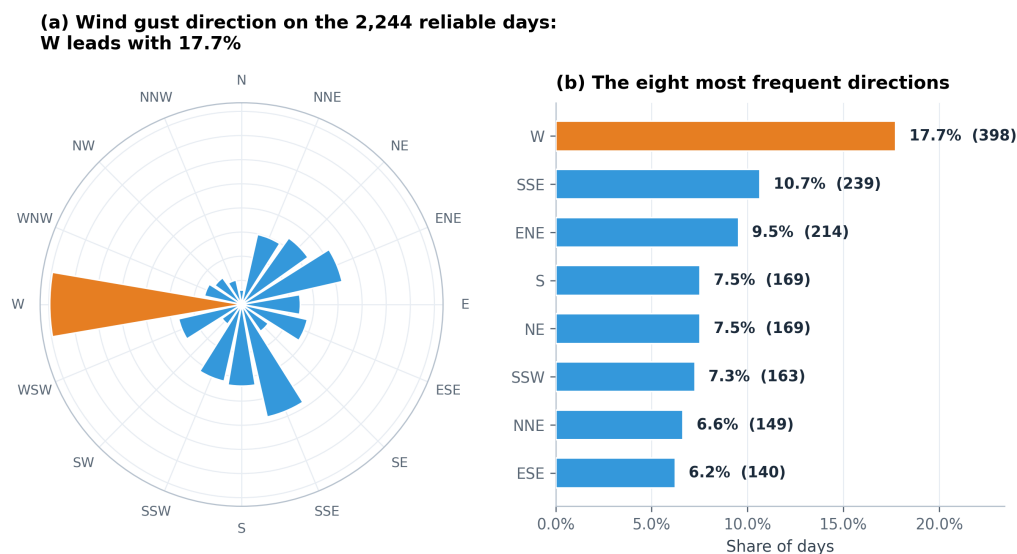


Figure 4. Wind gust direction, computed on the 2,244 reliable days. West leads with 17.7%, ahead of south-south-east at 10.7%. The 1,027 placeholder days are excluded: including them returns 43.6% for west, which is the backfill speaking rather than the site.

6.4 Trend

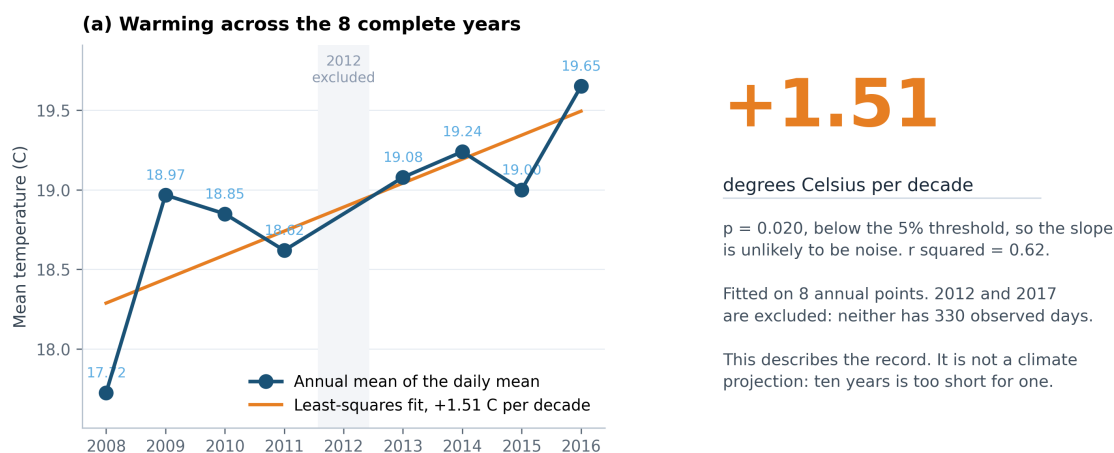


Figure 5. The warming trend, tested rather than asserted. The slope is $+1.51^\circ\text{C}$ per decade with $p = 0.020$, which is significant at the 5% level. The interval is nonetheless short, and the caption on the right states what the number does and does not support.

6.5 What moves with what

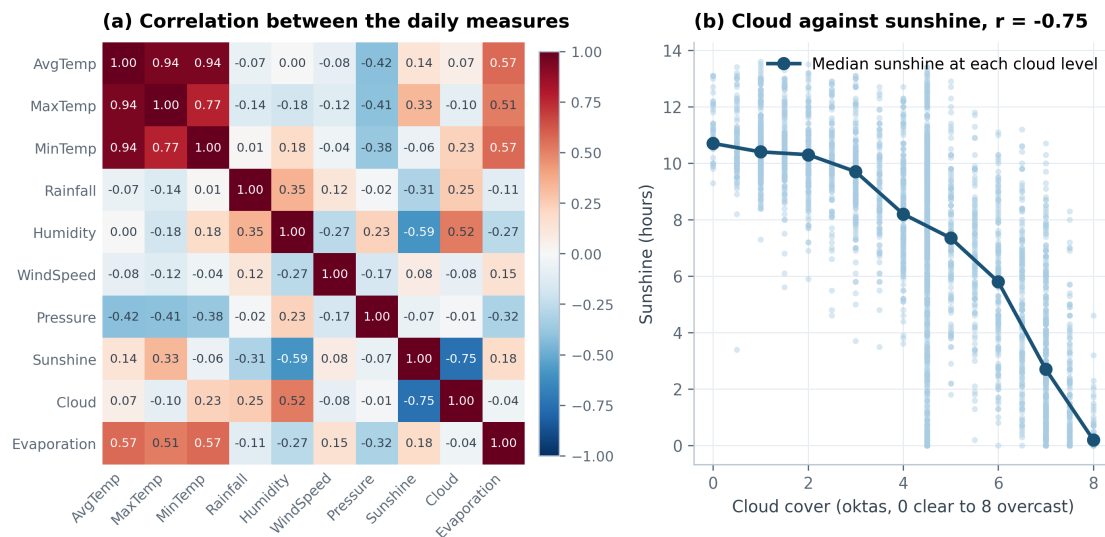


Figure 6. Relationships between the daily measures. The strongest association that is not mechanical is cloud cover against sunshine at $r = -0.75$: the two measure the same physical state from opposite directions. Temperature against evaporation at $+0.57$ is the strongest relationship between genuinely distinct quantities.

7. Insights

7.1 The record is not the calendar

162 calendar days carry no row, and whole months are absent. A monthly total therefore compares samples of unequal size, with a 29% spread between the best and worst covered month.

Here the correction does not change the ranking: June is the wettest month and September the driest on both the raw total and the per-day rate. **That is a result, not a formality.** It was established by computing both, and it means the seasonal conclusion can be stated with confidence rather than hope.

Recommendation. Publish rainfall as millimetres per observed day alongside the total, and show the observed-day count beside any monthly figure.

7.2 The seasonal signal is southern hemisphere

The warmest month is January and the coolest is July. Applying the northern convention would place the hottest quarter in winter and invert every seasonal statement in the study. The dashboard's humidity insight is exactly this error: humidity peaks in February, and February was read as winter.

Recommendation. State the hemisphere convention on the dashboard itself. It costs one line and removes an entire class of misreading.

7.3 Warming of 1.51 degrees per decade

The trend across the eight complete years is $+1.51^\circ\text{C}$ per decade with $p = 0.020$ and $r^2 = 0.62$. The slope is unlikely to be noise.

It is also fitted on eight annual points spanning nine years. A decade is short for a climate statement, and a single station cannot speak for a region.

Recommendation. Report the slope with its p-value and its sample size, never the slope alone. Describe it as a property of this record, not as a projection.

7.4 Rain and warmth are in opposition

Autumn delivers 4.00 mm per observed day and winter 3.63, against 2.31 in spring. June alone receives 3.4 times what September receives. The wet season and the warm season are in opposition, which is the single most useful fact in the record for anyone planning around it.

7.5 Cloud and sunshine measure the same state

Cloud cover and sunshine hours correlate at -0.75 . They describe the same physical state from opposite directions, so a model using both gains little from the second. The strongest relationship between genuinely distinct quantities is temperature against evaporation at $+0.57$.

Recommendation. Keep one of the pair in any predictive work, and prefer sunshine, which is measured on a continuous scale rather than in eight steps.

8. The Power BI dashboard

The dashboard presents the indicators, the seasonal profiles, the wind rose and the correlation panel on a single page.

It is cross-filtered throughout: selecting a month in the rainfall panel, a direction in the wind rose or a point in the correlation scatter filters every other visual at once, and the three slicers for year, month and season do the same. Every figure quoted in this report is the whole-record value, obtained with no selection active.



Figure 7. The Weather Analytics dashboard. Six indicator cards across the top, the temperature trend and monthly summary in the middle band, and the rainfall, humidity and wind panels below, with the key insights at the lower right. The capture was taken with a cross-filter selection active, which is why the cards read 20.24 °C and 224 rainy days rather than the whole-record values of 18.94 and 849 quoted in section 5: selecting any element filters every other visual on the page. The slicers for year, month and season work the same way.

9. Limitations and work in hand

9.1 What the data cannot support

No location. The file has no city, country or station column, so the spatial comparison the brief describes cannot be performed. Everything reported here describes one station.

Wind before 2011. 1,027 days carry the placeholder pair, including all of 2008 and 2009. Every wind figure in this report excludes them, and the exclusion is stated wherever a wind number appears.

162 absent days, clustered in 2010 to 2013 and removing whole months. Any figure not normalised for observed days is biased by them.

Eight years for a trend. The warming slope is statistically significant and still rests on eight annual points at one station. It describes this record and is not a climate projection.

RainToday is derived. It agrees with the rainfall column on every row, so it is not an independent observation and should not be treated as one.

Correlation, not causation. Nothing here establishes why the measures covary.

9.2 Corrections identified for the next dashboard revision

The insight panel of the dashboard was written before the analysis in sections 3 and 6 was complete. Four of its statements do not match the findings, and each has a straightforward fix.

Table 8. Items identified during this analysis, to be applied in the next revision

Item	Correction
Rainy Days card shows 1K	The count is 849. The card applies a thousands abbreviation to a three-digit number, which rounds up by 18%. Set the display unit to None.
“February recorded the highest rainfall”	June is the wettest month on both rankings, 5.90 mm per observed day against 4.32 in February. The Rainfall Analysis panel on the same page already shows June, so the two will be brought into agreement.
“Humidity peaks during winter months”	Humidity peaks in February at 66.7% and is lowest in September at 53.3%. February is summer under the southern-hemisphere convention established in section 4.
“Dominant wind direction: SE”	West leads with 17.7% on the reliable days; south-east accounts for 4.5%. The Wind Analysis panel displays E separately, and both will be recomputed on the reliable subset once the placeholder flag is added to the model.

Two statements on the page are confirmed by the analysis and need no change: summer is the warmest season, and the temperature trend is upward. The report quantifies the second at +1.51 °C per decade, which the dashboard can now state rather than imply.

The underlying cause of three of the four. The placeholder in the wind record and the hemisphere convention were both established in this report, after the dashboard was built. Adding the reliability flag and the season field to the Power BI model resolves the wind and humidity items together.

Deliverables

File	Content
Weather_Analysis.ipynb	Commented notebook, executed
report.pdf	The present report
INSIGHTS.md	Insights and recommendations
data/weather_clean.csv	Cleaned dataset, 3,271 rows
analysis.py	The code, commented
figures/	Six analysis figures
Weather_Dashboard.pbix	The Power BI dashboard